# Quiet Climate Communication: Beauty and Poetry

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## Abstract

Societal understanding of climate change, sustaining ecosystems, and preventing decay of existing environmental balances are critical to maintaining a habitable global environment. Solutions to these pressing issues require collaboration between scientific and artistic communities. Motivating the implementation of solutions requires the humanities to join the science community to usher in trust of the science, understanding of our climate and engagement by society. Visual and literary communication methods have been used by civilizations as far back as we have documentation. These traditions speak both factually and emotionally via artistic voices embedded with an understanding of the science. The three such collaborations are presented here.

## Keywords

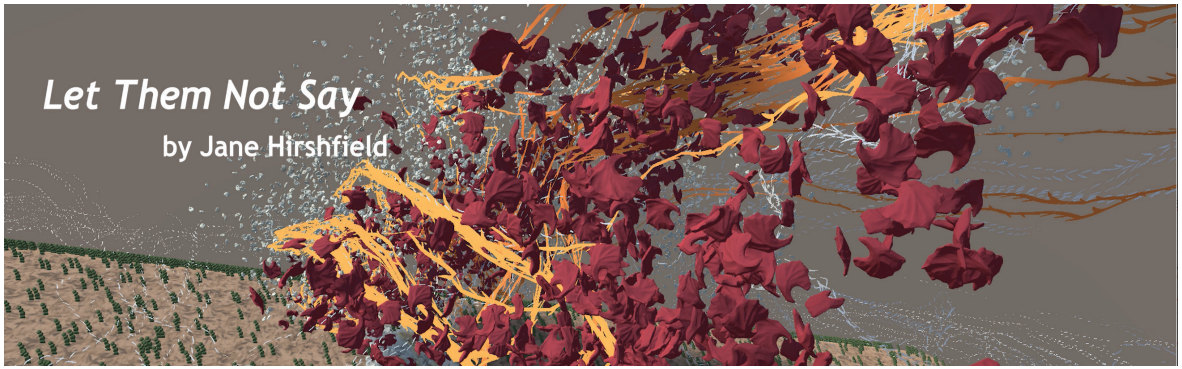
Climate communication, poetry, art-science collaboration, data visualization, wildfire, Arctic field studies, Greenland's ice sheet

## Introduction

Scientists have identified effects of anthropogenic climate change, including sea level rise and flooding, habitat loss, decreases in biodiversity, extinction and threats to food supplies [1]. They have argued for sustaining ecosystems and preventing further decay of environmental balances in order to maintain a habitable global environment [2].

However, Hulme, a climate scientist, noted that crafting increasingly consensual reports of the scientific knowledge, or leveraging more engineering and technology, will alone never open up pathways from the research to the public imagination or execution of policy.
He makes the case for elevating storytelling alongside the fact-finding in order to help society understand and respond to climate change. Intellectual understanding as well as an emotional connection is required to motivate concrete action that includes lifestyle changes [2, 3].
This portfolio presents collaborations between Samsel and three poets working toward that goal..

Figure 1. The opening of *Let Them Not Say,* blending poetry and visualization about wildfire.



## Collaboration Foundations

Science and art are often characterized as opposing modes of inquiry—one concerned with objective fact, the other with subjective experience. However they are both means through which we explore and build understanding of our world.

Here we present a transdisciplinary practice in which science and art both contribute paths through the content facilitating engagement designed to reach a wide range of audiences.

The three collaborations presented here focus on accurate representation the science that also creates a human connection to nature and the changes occurring on the planet. Rather than abstracting from the science, the artistic language depicts and augments the scientific content thus creating new points of entry.

Samsel's practice involves long term collaborations where mutually beneficial outcomes are defined. AN early understanding of the investment and the outcomes sets the foundation for the scientists being willing to invest necessary the time and embed her in their work. This approach enables collaborations to move beyond illustration and incorporate other dimensions for deeper involvement.

The role of the poetry is to provide entry points that visuals alone can not convey. Sources include: in-depth interviews with the scientists; field blogs written by the science teams; field and lab notes; and publications of the work and more.

Working with Benjamin Keisling and colleagues who are studying the Greenland Ice Sheet [5], Samsel needed to understand not only their findings, but: the scientific questions driving the research; the research methods; the experiences and challenges of working in extreme environments; the data collection methods and analysis; and much more.

The imagery and poetry emerges from these sources as well as research documentation. In all three projects that scientists reviewed all of the content prior to inclusion in the final work. Each scientist chose varying levels of involvement. Keisling was involved in all stages for the process from idea generation to display of the final work. Cathy Wilson, science lead on the NGEE - The Acrtic [6], Climate Prisms project, spent endless hours with the artistic team discussing not just the nuts and bolts of the science but more nuanced aspects that enabled cross-connect between the science and the metaphors. She personally reviewed all 800 pieces of final content developed

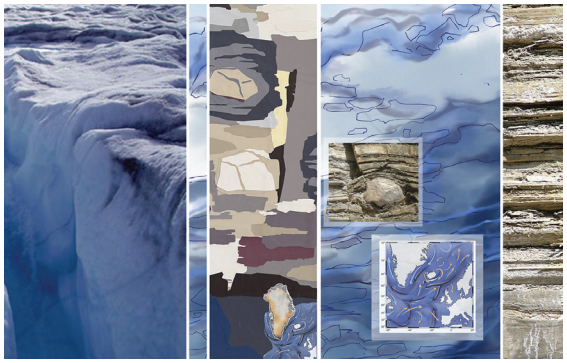


Figure 2. Content from *Bedrock,* an earlier collaborative work between Keisling and Samsel, did not contain poetry, requiring viewers to read lengthy descriptions.

In *Let Them Not Say*, data was generated by the fire simulation model [7] developed by Rob Linn, who Samsel had worked with extensively. Here the data is presented in hand-made artistic forms [9], as shown in Figure 1.

Science and art are both means through which we understand the world and our place within it. Science provides evidence, explanation, and precision; art and poetry provide metaphor, emotional resonance, and the capacity to engage with ideas that may otherwise feel distant or overwhelming. This distinction is especially important in communicating climate change. Factual accuracy is essential, but facts alone seldom foster a lasting personal connection. Poetry offers a complementary mode of understanding, one that connects scientific knowledge to human experience.

*Painting is poetry which is seen and not heard.*

*Poetry is a painting which is heard and not seen.*

Leonardo da Vinci

## Poetry for Climate Communication

Combining poetry with visual imagery offers a mode of climate communication that bridges empirical science and lived human experience in ways that data alone often cannot. Research in environmental communication and ecopoetics demonstrates that poetry's metaphorical precision and emotional resonance can personalize abstract climate processes and make scientific knowledge more accessible and meaningful [3]. Theoretical work on climate geopoetics. further argues that poetic language provides a critical framework for situating environmental data within cultural, ethical, and affective contexts [4]. When paired with visual elements such as scientific visualizations, interactive exhibits, or environmental imagery, poetry becomes a mediating lens through which audiences can simultaneously see and feel climate change, aligning cognition with emotion [8]. Empirical studies show that arts-based approaches, including poetry, are particularly effective in engaging under-served or non-expert audiences with environmental issues by fostering reflection, empathy, and dialogue rather than persuasion alone [3]. Together, poetry and visual imagery support a form of climate communication that connects facts to values, enhancing understanding of both the scientific realities of climate change and the human stakes of action or inaction.

Figure 3. Woven Gifts, a mural about the diverse cultures that have lived and worked in Yonkers New York, building it into the melting pot of cultures that it is today. Mural, 140' x 20'. By Francesca Samsel, 2007

Samsel, a mural painter before turning to visualization, has long incorporated poetry into her work, providing another inroad for viewers. In Woven Gifts, Figure 3 , there are poems by Emily Dickenson, who lived in Yonkers, Maya Angelou, Francis Bacon and others.

# Climate Prisms: The Arctic, Poetry by Michael G. Smith

Poetry provides a context linking cause and consequence, fact and feeling. Through the precision of language—and its capacity to both relax and stretch meaning—poetic metaphor becomes the exhibit's connective tissue, melding scientific knowledge with human values. In this work, poet Michael G. Smith interprets observations of climate change in the Arctic and the complex science underlying it, with particular attention to the short- and long-term repercussions of human inaction and the resulting environmental degradation.

The poetry pieces shown here are part of Climate Prisms: The Arctic, which is an interactive museum exhibit Samsel created in collaboration with NGEE Arctic researchers from Los Alamos National Laboratory [10]. Commissioned by The Bradbury Science Museum, the exhibit integrates art, poetry, and scientific visualization to bridge empirical data and lived experience.

Smith collaborated with Samsel on a series of poems based on the NGEE Arctic scientist's experiences, a selection of which are presented here in Figures 4 - 6. Specifically he drew on the scientists blogs of the field time experiences. Smith's purpose was to invite viewers to encounter climate science not only as information, but as consequence, responsibility, and shared future.

## Thinking About the Data

There is are those times when you must
walk away from the laptop or white board
scribbled with notes and equations and step
past the door into the winter that seemingly
goes on forever. You've stopped being
in the world of hard numbers.
In that world, life has no meaning.
But out here the snow is windblown,
and each boot fall requires a firm push
to break through the crust. And out here
your lone footprints mark the passage
of someone who imagined and dared.

Figure 4, Thinking About Data, connects us to the researchers themselves, via an internal voice, and highlights internal struggles recognizable to us all.

## Feedback Loop

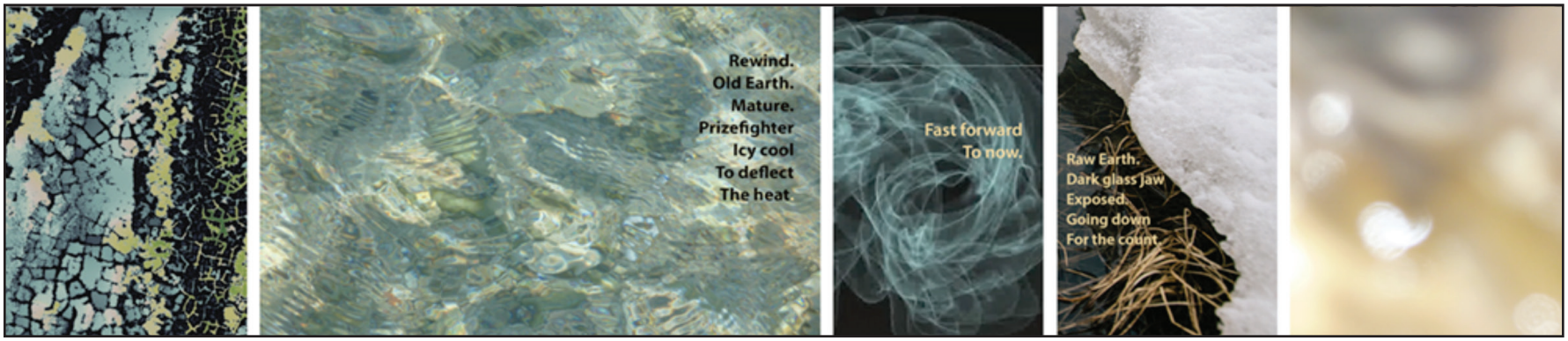



Figure 5 *Feedback Loop* speaks to the role of water in all facets of the climate. The center image is a visualization of water vapor forced into the stratosphere by an asteroid impacting the ocean. Watching the animated visualization of the water vapor is as mesmerizing as watching the patterns and powers of nature.

*"Our primary goal for using all the assorted art in Climate Prisms was to find a way "in," so that viewers who were apt to want to ignore studies about or evidence of climate change would be drawn in an unexpected, affective way.*

*The poetry in Climate Prisms, overlain on photos or art, gives a richer experience of the visuals presented, supplementing them and slowing down how the offering would be consumed. In this way it seemed that the poetry provided a more personal, intimate way, through reading slowly and thoughtfully which poetry requires, to engage with our message."*

Linda Deck
Executive Director
Bradbury Science Museum
Los Alamos, New Mexico

*Feedback Loop,*

*Feedback Loop* shown in Figure 5 and the poem itself on the right, connects the physical environment to its abstraction representation, a lidar visualization of the topography of the Arctic tundra, linking the abstract scientific recording to the physical environment represented. Visualizations, photographs, drawing, and poetry were all derived from NGEE - Arctic documentation [6]. The Abstract regions of the work are intentional, designed to provide a space for the viewer to add their own interpretations from their own life experiences.

Rewind

Old Earth.

Mature.

Prizefighter

Icy cool

To deflect

The heat.

Fast forward

To now.

Raw Earth.

Dark glass jaw

Exposed.

Going down

For the count.

## Tipping Point

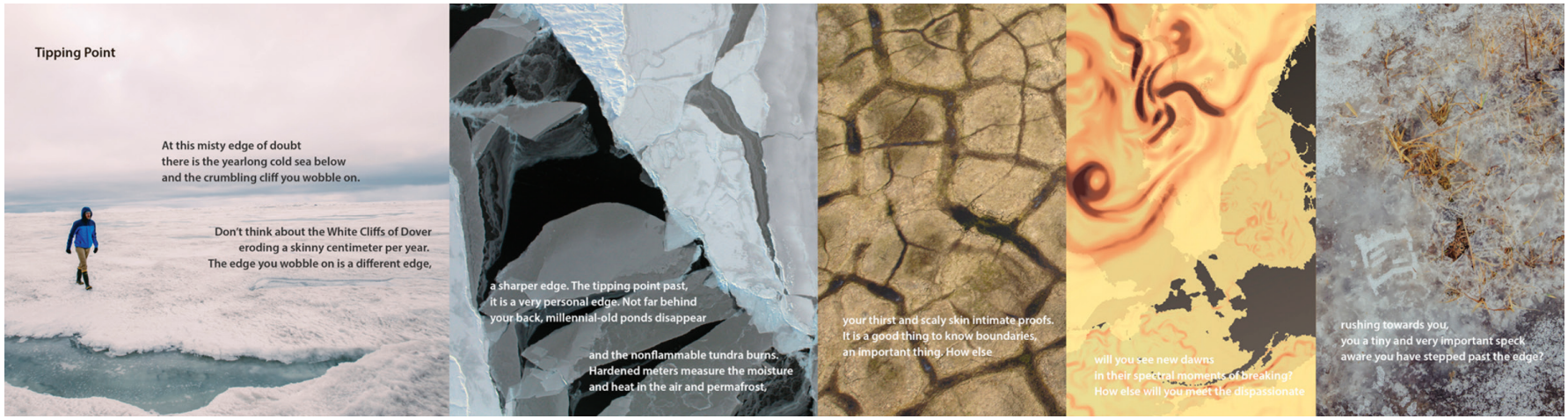



Figure 6. *Tipping Point* - images from left to right: researcher on the tundra; aerial view of colliding sea-ice; drained dry tundra; visualization of ocean. turbulence – MPAS-Ocean which is related to erosion; melting tundra ice, exposing last season's growth. https://vimeo.com/138889021?fl

### Tipping Point

At this misty edge of doubt
there is the yearlong cold sea below
and the crumbling cliff you wobble on.

Don't think about the White Cliffs of Dover
eroding a skinny centimeter per year.
The edge you wobble on is a different edge,

a sharper edge. The tipping point past,
it is a very personal edge. Not far behind
you back, millennial-old ponds disappear

and the nonflammable tundra burns.
Hardened meters measure the moisture
and heat in the air and permafrost,

your thirst and scaly skin intimate proofs.
It is a good thing to know boundaries,
an important thing. How else

will you see new dawns
in their spectral moments of breaking?
How else will you meet the dispassionate

rushing towards you,
you a tiny and very important speck
aware you have stepped past the edge?

Michael G. Smith

*"I am an outdoor enthusiast, a backpacker and a snowshoer. A good chunk of my motivation and enthusiasm for doing so is the immediate and absolute connection to the environments I'm in, whether it be the Grand Canyon and old Growth Forest or hiking in Denali National Park. I find such immersions bring out the best imagery in my writing, and I know that immersing myself in northern environments such as Iceland that are going to be the hardest hit by climate change will result in deeply moving and connecting poems that hopefully motivate people to action."*

Michael G. Smith
Poet

## Between Melt and Memory

Driving this collaboration between Samsel and Keisling is a concern for the accessibility of climate science. The processes by which knowledge is created—sampling, modeling, simulation—often remain opaque to the public, even as their implications shape the future of habitability. *Between Melt and Memory* melds scientific artifacts with artistic transformation to provide pathways into this complexity. It does not ask viewers to master the data. Instead, it offers entry through curiosity and wonder, acknowledging that not all facts must be present to begin contemplating responsibility and change.

At each of three sites, scientific analysis of ice, sediment and rock reveals the last time these ancient materials saw the sun – meaning the ice was smaller years ago; and at the Summit, one million years ago. Closest to the coast at Prudhoe Dome, the relatively recent exposure of these materials tells us with a high degree of certainty that the ice sheet receded by ~10% in response to limited high-latitude warmth [5]. As we traverse deeper inland to the other sites, our knowledge becomes less certain—yet these sites tell of times when the ice sheet was much smaller or even absent. If warming continues, such states become plausible. That risk motivates continued scientific innovation and the application of high-performance computing to unravel the mysteries these ancient archives encode.

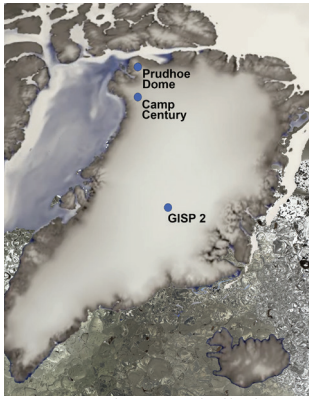



Figure 7 The core drilled research sites in Greenland.

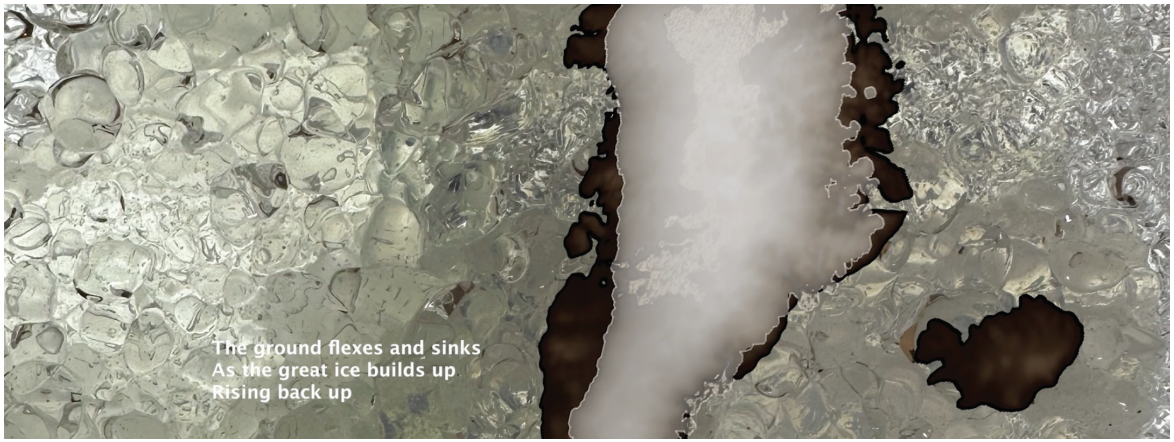
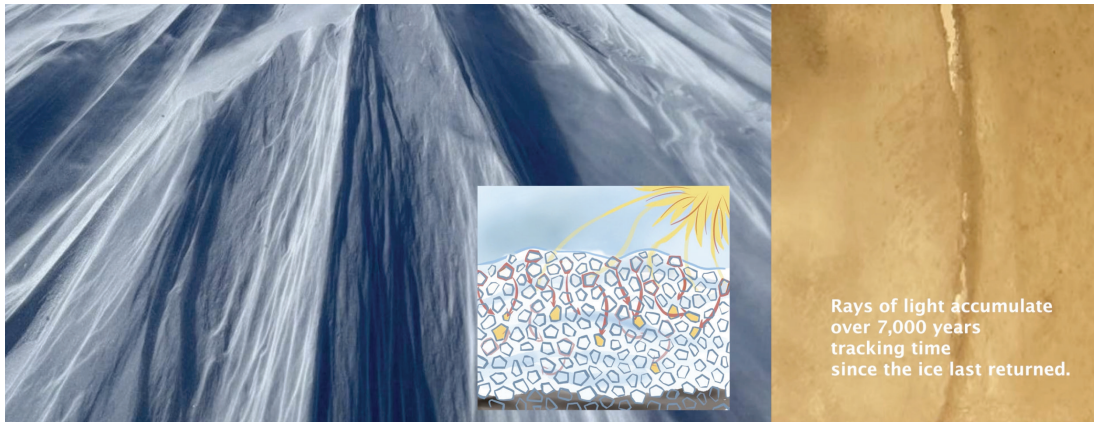


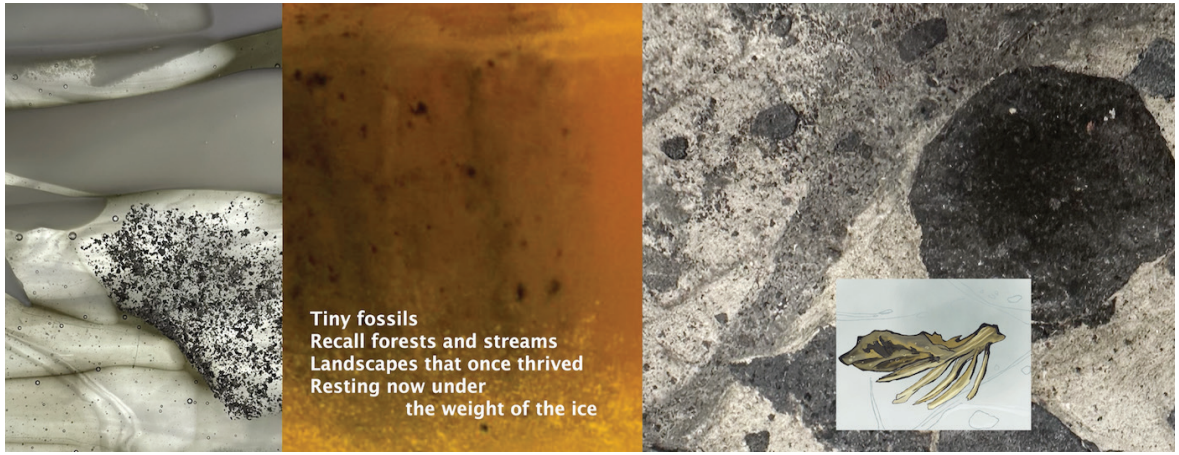

Figure 8 About the movement of the bedrock, right.

Figure 9 On Prudhoe Dome, optically stimulated luminance, below.




The ground Flexes and sinks
As the great ice builds up
Rising back up

Rays of light accumulate
over 7,000 years
tracking time
Since the ice last returned.



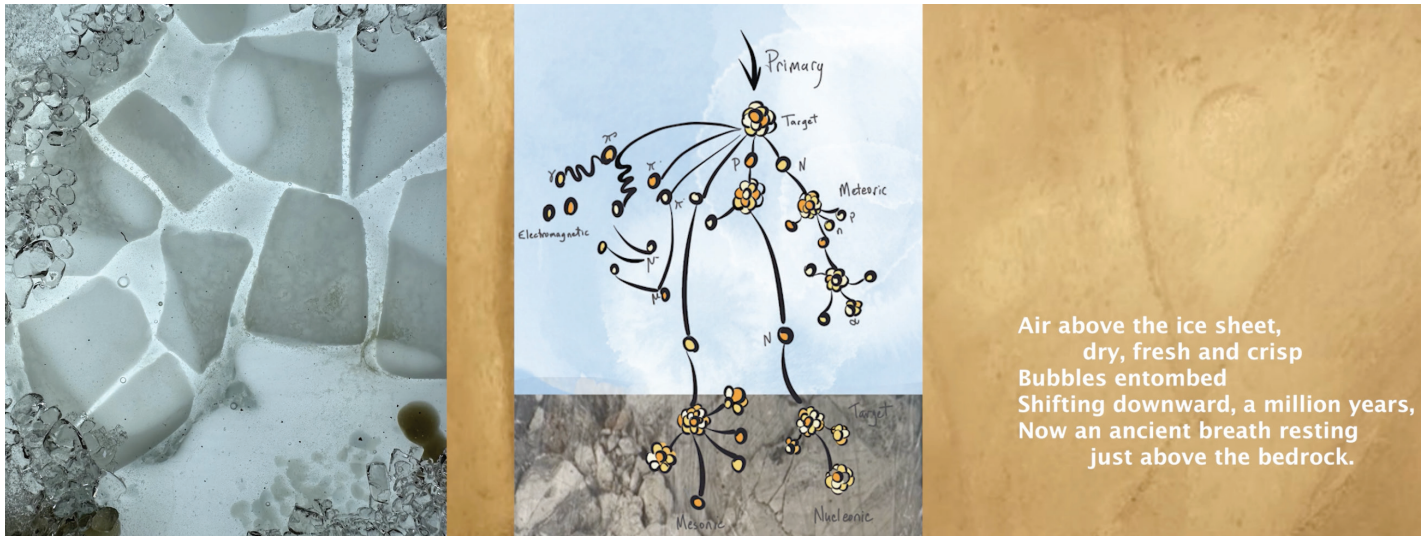

Tiny fossils

Recall forests and streams

Landscapes that once thrived

Resting now under the weight of the ice.

Figures 10 & 11. Panels from Between Melt and Memory, about the work and scientific methods at Camp Century.




*Between Melt and Memory* traces the spine of Greenland's ice sheet, from the thinning, melting margins near the ocean to its two-mile-thick summit some 700 kilometers inland. Scientific methods reveal that this immense body of ice has been far smaller in the past, responding to prolonged warmth over millennia. Subglacial matter becomes data, becomes model, becomes image, and finally becomes felt experience. The work translates isotopic records and computational simulations into visual and tactile languages that mediate between scientific models, display systems, and public perception.

Air above the ice sheet, dry, fresh and crisp

Bubbles entombed

Shifting downward over a million years,

Now an ancient breath resting

## Wildfire and a Climate Poet

A third thread in this exploration of poetry and climate communication emerged through Samsel's collaboration with poet Jane Hirshfield, whose work has long examined humanity's relationship with the natural world and who founded the Poets for Science initiative. Their partnership began unexpectedly following an introduction at a National Academy of Sciences event. At the time, Samsel was creating fused-glass and ceramic interpretations of Greenland ice cores and shared images of the work with Hirshfield. In response, Hirshfield sent an unpublished draft of To Sand—An Assay, beginning a creative dialogue about the ways poetry and visual imagery might together communicate scientific understanding. As with Samsel's earlier collaborations, a single exchange grew into an ongoing creative relationship.

The collaboration expanded when Samsel paired Hirshfield's widely recognized climate poem *Let Them Not Say* with visualizations derived from wildfire simulations generated using the HiGRAD/FIRETEC model developed by Rod Linn and colleagues [7]. The scientific imagery was not intended to illustrate a particular wildfire, just as the poem itself does not describe a single event. Instead, both speak to a shared human experience of wildfire in an era of climate change. Together, they demonstrate how scientific visualization and poetry can address the same subject from complementary perspectives—one grounded in measurable processes, the other in meaning, memory, and human responsibility. Neither serves as an illustration of the other; rather, each enriches the other's capacity to communicate the realities of a changing climate.

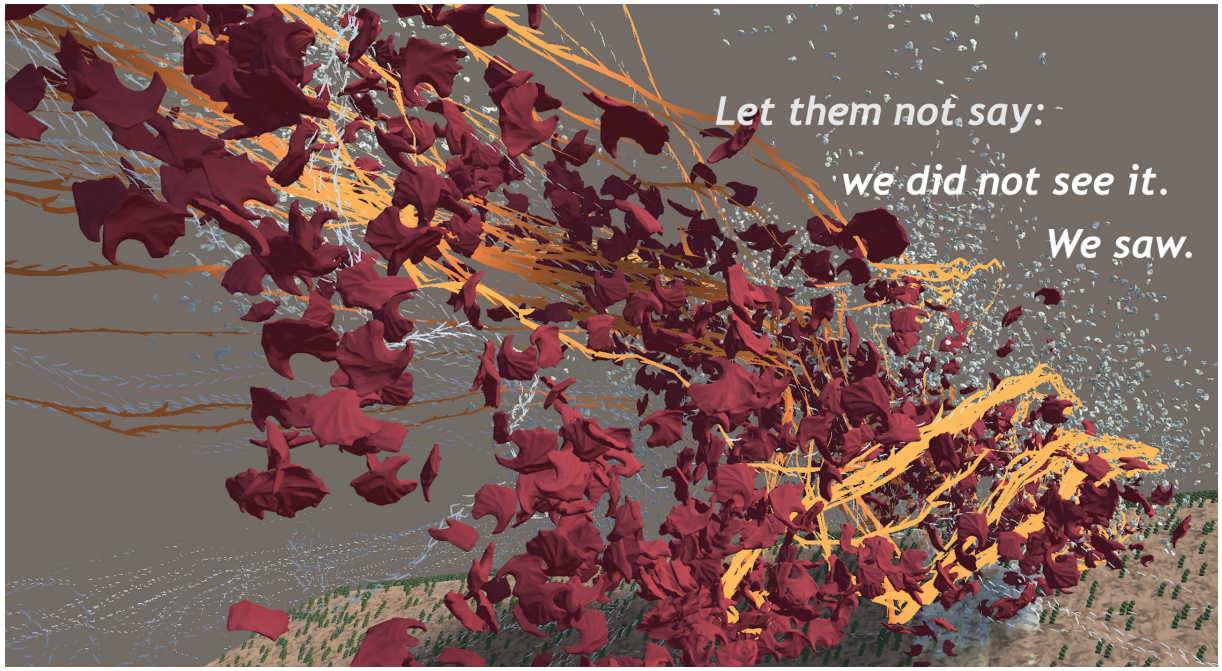
## About Jane Hirshfield

Jane Hirshfield is an internationally acclaimed poet and essayist whose work addresses climate change, extinction, and humanity's ethical relationship with the natural world. Her poetry and essays bring environmental urgency into intimate, reflective language that invites attention, empathy, and responsibility.

A founding voice of Poets for Science, Hirshfield has helped mobilize writers in public support of science and climate action, demonstrating how poetry can engage civic life and deepen environmental understanding. She has presented poetry and spoken on the role of art and responsibility at Nobel Prize–related events and other international forums, where her work highlights how imagination and beauty can shape meaningful responses to planetary crisis.

The original video of *Let Them Not Say* is available at https://vimeo.com/1129924529. This work was orignally shown at The International Conference on High Performance Computing in 2025, in The Art of HPC Exhibit.

Figures 12 Hirshfield and Samsel's first collaboration - *To Sand: An Assay* originally published in Leonardo, MIT Press, 2026 [11].




*Let Them Not Say*
by Jane Hirshfield

Let them not say:
we did not see it.
We saw.

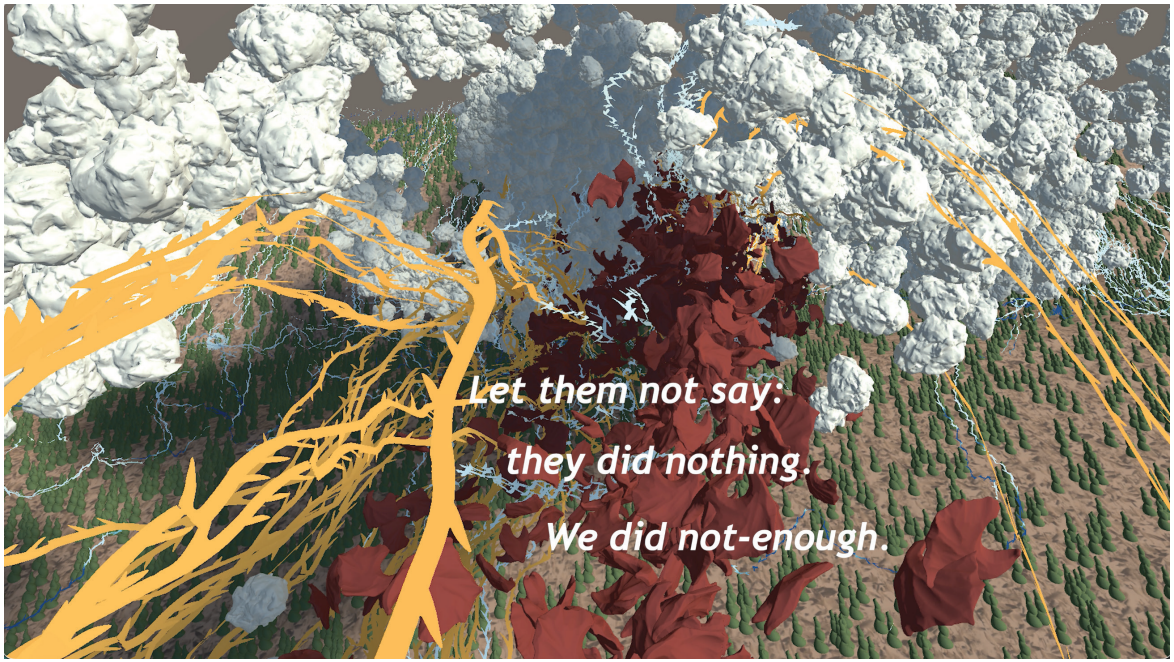
Let them not say:
we did not hear it:
We heard.

Let them not say:
they did not taste it.
We ate, we trembled.

Figures 13, 14 & 15. *Let Them Not Say*

Let them not say:
it was not spoken, not written:
We spoke,
we witnessed with voices and hands.

Figures 16 & 17. *Let Them Not Say*




Let them not say:
they did nothing:
We did not-enough.

Figure 18. *Let Them Not Say*

Let them say,
as they must say something:
A kerosene beauty
It burned.

Figure 19. *Let Them Not Say*

Let them say we
warmed ourselves by it,
read by its light,
praised,
and it burned.

Jane Hirshfield
poet

## Wildfire Simulation Data

The visualizations in "Let Them Not Say" uses data from the Higrad/Firetec Wildfire Simulations, developed by Rod Linn [7]. Visualizations of the same data created in ParaView are shown in Figures 20 and 21. These visualizations are complicated because of the many factors involved in how wildfires develop. The simulations shown here are created by Samsel with the intent of providing scientists with the information they need to understand and predict wildfire behaviors.

Wildfire behavior is a result of complex nonlinear interactions between a variety of physical processes and chemical reactions. Understanding the coupling between these components and the ways in which their interactions are influenced by their surrounding environment (e.g. winds, fuels, and topography) is a challenging undertaking. This simulation examines vorticity-driven lateral spread (VLS) in wildfires of mountain and canyon topographies.

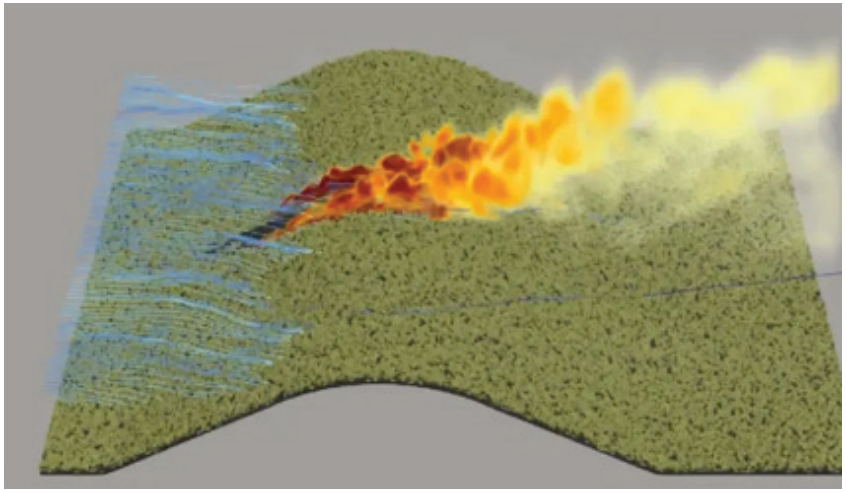
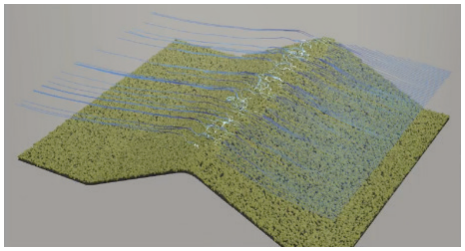
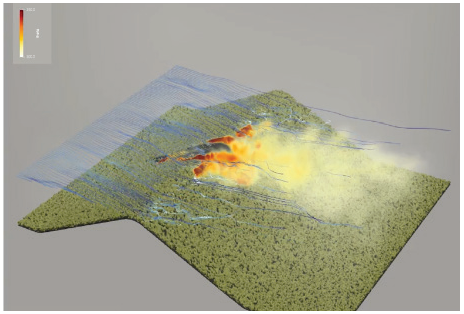


Figure 20. Wildfire simulation using the same data as the images accompanying Let Them Not Say.

These nonlinear interactions, and corresponding range of scales of processes influencing wildfires, contribute to surrounding environment (e.g. winds, fuels, and topography).

The visualizations made for *Let Them Not Say* present the same exact data, but in a language designed for its expressive capacity. The visualizations are built using Artifact-Based Rendering which enables artist-created glyphs, textures lines and colormaps to be applied to data [9].





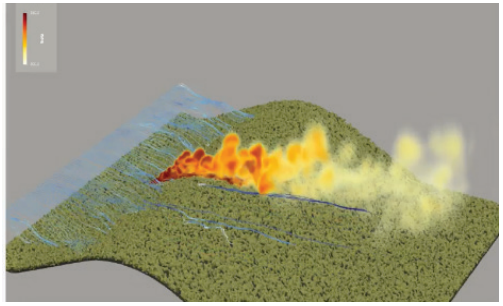
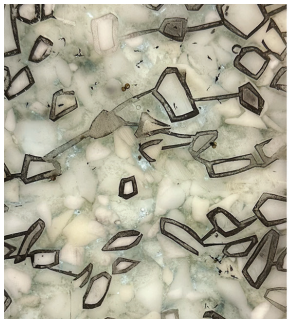


Figure 21 Visualizations from the ensemble of wildfire data.

## Future Work

Work underway, furthering the Keisling / Samsel collaboration involves physicalization of the core samples, rendered in fused glass. Here, shown in Figure 22, frames 1 and 4, the physical material, the fused glass, linking both to ice and geology, provide a magic that comes from the endless nuanced complexity found in the pieces as well as in nature.

Unlike earlier work the glass renditions of the cores, the textures and patterns of specific cores at specific locations but they are artistic interpretations not actual replicas. However, Samsel and Keisling are experimenting with combining photographs of the cores in Figure 23. Figure 22 shows textured



with photographs of the glass as shown and sand blasted glass thus adding the sense of touch to the work. There are plans for sanding blasting the poetic excepts into the glass expanding the poetry into a tactile experience.

In a continuing effort to collaborate with poets, Samsel has reached out to Poets for Science, an active organization founded by Jane Hirshfield.

## Conclusion

Rather than using science as subject matter for artistic interpretation, the collaboration treats scientific practice itself as a creative partner. The research, data, and field experience provide the structure from which the artistic work emerges, allowing poetry and visualization to reveal dimensions of the science that are difficult to convey through scientific discourse alone.

Figure 23. Visual poetry, comprised of images of fused glass, science illustration, a photograph of the drill used at Camp Century in the 1960's, images of core samples from Camp Century and a visualization of Greenland's ice sheet cover, using Keisling's ice sheet model.

## Acknowledgements

All of the projects presented here required the efforts of many collaborators.

*Climate Prisms* had teams of support, both scientific, technical and financial starting all organized by the Bradbury Museum of Science and their Director, Linda Deck. All of the poetry within the work was written by Michael G. Smith. The technical efforts to produce the interactive exhibit were lead by Bruce Campbell and Greg Abram. Cathy Wilson, the lead scientist of the NGEE team and Joe Rowland and Brent Newman, spent countless patient hours with us explaining the science the experiments, the reasoning for the work and the physics underlying the ongoing changes in the geological formations of the region.

*Between Melt and Memory* draws directly from the research of Benjamin Keisling, UTIG UT Austin. The visualizations were supported by Greg Abram, TACC, UT Austin.

*Let Them Not Say* stands on the poetry of Jane Hirshfield (www. https://www.poetryfoundation.org/poets/jane-hirshfield ) Jane Hirshfield is the founder of Poets for Science.

The HiGRAD Wildfire simulation model used in *Let Them Not Say* was developed by Rod Linn, LANL [11].

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Figure 24. Jane Hirshfield presenting at a Nobel Prize climate summit